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Rock magnetic and magnetic fabric variations across three thin alkaline dykes from Sunnhordland, Western Norway; influence of initial mineralogy and secondary chemical alterations

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SUMMARY

Profiles of natural remanent magnetization (NRM), rock magnetic properties and anisotropy of magnetic susceptibility (AMS) across three thin (26 to 109 cm) alkaline dykes from Sunnhordland, Western Norway are presented.

The magnetic remanence is carried by pseudo-single-domain (PSD) low-temperature oxidized titanomagnetite, with circa 6 per cent of cations substituted by aluminium. Systematic variations in magnetic properties across the dykes are caused by differences in titanomagnetite concentration as well as grain size. Low-temperature alteration and laboratory oxidation by heating in air to a maximum temperature of 600 °C, do not homogenize the magnetic properties across the dykes. Fine-grained regions near dyke margins carry the most stable remanent magnetization, but are also most amenable to secondary chemical remanent magnetization (CRM) resetting.

Between dyke variations in magnetic properties are larger than within dyke variations. A stable primary remanence is found in two of the dykes, while the third is totally reset by a soft viscous remanent magnetization (VRM), despite similar optical grain sizes, magnetic mineralogy and oxidation state. The difference is attributed to a lower titanium content in the latter dyke, preventing effective partitioning of magnetic grains during low-temperature alteration.

Laboratory heating affects anisotropy variations across the dykes, resulting in more consistent axis orientations within each dyke, and symmetric edge-to-edge variations in the degree of anisotropy. AMS directions are different from those in the country rock, and vary systematically across the thinnest dyke before heating.

Key words: dykes, magnetic fabric, palaeomagnetism, rock magnetism.

1 INTRODUCTION

Systematic variations of magnetic properties across igneous structures with differential cooling rates are well documented. Studies of individual lava flows, dykes and pillow lavas reveal patterns relating to initial grain-size variations as well as differential secondary chemical alterations of the titanomagnetite phases.

In basaltic lava flows extensive hydrothermal alteration tends to obscure the initial grain-size dependence of magnetic properties and gives rise to lateral variations (Ade Hall *et al.* 1968a). Cooling rates in thick flows may be slow enough to allow deuteric oxidation of titanomagnetite grains (Watkins & Haggerty 1965; Ade Hall *et al.* 1968a, 1968b; Wilson, Haggerty & Watkins 1968; Lawley & Ade-Hall 1971), making observed magnetic profiles complicated functions of a generally high-oxidation state as well as initial grain size.

Profiles across individual pillow basalts are influenced by hydrothermal oxidation, but variations in magnetic properties due to original cooling rate differences have been observed both for young *in situ* oceanic pillow basalts (Watkins, Paster & Ade-Hall 1970; Marshall & Cox 1971; Soroka & Beske-Diehl 1984), and in an Ordovician ophiolite complex (Trench *et al.* 1990).

Systematic rock magnetic studies on dykes are few. Petersen (1976) studied variations across a 180 cm thick Holocene dyke from Vesuvius, containing relatively unaltered homogeneous titanomagnetite [Curie temperatures (T_c) generally below 300 °C]. Maximum values near dyke margins of remanent intensity (J_{NRM}) coincided with minimum values of susceptibility (K). Between the margins and centre of the dyke, the ratio of saturation remanence to saturation magnetization (J_{rs}/J_s) fell from above 0.3 to below 0.1, indicating a transition from PSD to true

multidomain (MD) behaviour for titanomagnetite (Dunlop 1981). These observations are in qualitative agreement with the observed increase in average titanomagnetite grain size from less than 10 μm at the margins to more than 120 μm towards the centre of the dyke.

Smith & Prevot (1977) studied a much thicker (20 m) basaltic dyke of Matuyama age from the Massif Central, France. This dyke has undergone extensive deuteric oxidation and some low temperature alteration, giving high T_C's of around 550 °C across most of the dyke. Despite this, the same grain-size dependent trends were observed, in addition to a zone of high susceptibility and low remanent stability closest to dyke margins, attributed to the presence of very small superparamagnetic (SP) grains.

Schoenharting & Hall (1982) found little systematic variation in susceptibility across profiles of numerous dykes within a 2 km long drill core of Miocene Icelandic basalts. They attributed this to an increase in deuteric oxidation towards dyke contacts balancing grain-size effects.

The present study focuses on three thin alkaline dykes (26, 65 and 109 cm wide respectively at the sampling localities) of Jurassic or greater age. Spatial variations in AMS across the dykes are studied, as well as detailed profiles of remanent stability, bulk magnetic properties and hysteresis parameters. The effects of heat-induced laboratory oxidation were investigated, and inter and intradyke relationships are discussed.

2 GEOLOGICAL CONTEXT AND SAMPLING

The three studied dykes, named 'SA', 'UH' and 'ST' for the 25, 65 and 109 cm profiles respectively, belong to a suite of more than 60 alkaline dykes in Sunnhordland, Western Norway (59.8°N, 5.7°E). They typically strike in a NNW direction with sharp vertical contacts with the gneissic country rock (Færseth, Macintyre & Naterstad 1976). K-Ar measurements (Færseth *et al.* 1976) give ages of 216, 274 and 254 Ma for the SA, UH and ST dykes respectively. Løvlie (1981), who carried out a palaeomagnetic investigation on 18 of the dykes, found no systematic differences in remanence directions between dykes with differing K-Ar ages, leading him to suggest that the radiometric ages are unreliable. Half of the measured dykes carried a stable single component remanence. The mean direction ($D = 48$, $I = 49$, $\alpha_{95} = 5°$) in stable dykes is discordant with the apparent polar wander path for Europe, but may be made to coincide with the Triassic/Jurassic palaeopole position by an inferred tilt correction of 15° (Løvlie 1981). Remanence in the nine remaining dykes was highly unstable, but a poorly defined reverse component was inferred to be of Permian age.

The ST and UH dykes were both sampled with a field drill along a profile across each of the dykes. Individual cores ($d = 1.9$ cm) were between 2.5 and 3 cm apart, and were oriented with a magnetic compass. Additional cores were drilled adjacent to and cutting through the contacts with the country rock. A total of 44 cores yielded 68 specimens from the ST dyke, and 30 cores yielded 46 specimens from the UH dyke. In addition small diameter ($d = 0.8$ cm) cores were obtained from respectively 12 (ST) and 10 (UH) of the larger cores for hysteresis measurements.

Cores from the thinnest (SA) dyke were obtained from a single oriented hand sample spanning the total width of the dyke. Two parallel profiles less than 3 cm apart yielded a total of 20 cores or 34 specimens, which were combined to form a single profile during the subsequent data analysis. An additional profile of 24 small diameter cores for hysteresis measurements was drilled adjacent to the larger cores.

3 MAGNETIC MINERALOGY

The magnetic minerals were studied by means of reflected light microscopy, scanning electron microscope (SEM) and thermomagnetic analysis.

Titanomagnetite (TM) (Fig. 1) is the dominant magnetic phase in the three dykes, showing an increase in maximum grain size from 5 μm near contacts with the country rock to 30 μm (UH dyke) and 100 μm (SA and ST dykes) respectively in the centre. No exsolution lamellae of ilmenite were detected, indicating cooling rates too rapid for deuteric oxidation. Curvilinear shrinkage cracks indicate low-temperature alteration of TM, and formation of cation-deficient phases. The ST dyke is apparently the most altered, with red haematite staining of the matrix as well as secondary opaque phases along grain boundaries, in particular close to dyke contacts.

Non-magnetic pyrite is present in all three dykes, but does not show the same spatial size variations as the TM phases. There is a marked inverse relationship between pyrite and TM concentration in the UH dyke, with pyrite being the dominant phase near dyke contacts, and TM dominating the central parts.

Energy dispersive analyses (EDA) were performed on specimens from the central parts of each dyke. Bulk TM compositions (Table 1) are consistent within each separate dyke, and may be summarized as $Fe_{2.28}Ti_{0.55}Al_{0.17}O_4$ (ATM 55/17) in the SA dyke, $Fe_{2.22}Ti_{0.57}Al_{0.21}O_4$ (ATM 57/21) in the UH dyke, and $Fe_{2.52}Ti_{0.29}Al_{0.19}O_4$ (ATM 29/19) in the ST dyke, neglecting the cation deficiency caused by low-temperature oxidation. Aluminium is present in large enough amounts to have a significant effect on magnetic properties (Özdemir & O'Reilly 1981; O'Reilly 1984).

TM grain dimensions close to dyke contacts are comparable to the electron beam width, making EDA measurements less reliable due to contamination from neighbouring grains. No results were obtained near contacts in the UH dyke. Analyses near margins of the SA dyke indicate the same bulk TM composition as in the dyke centre within experimental error. In contrast, the opaque cubic phase less than 2 cm from the margins of the ST dyke seems to be pure magnetite, which may, however, be of secondary origin.

Thermomagnetic analysis (Fig. 2) was performed in air on a horizontal translation balance, with a heating/cooling

Table 1. EDA element analyses of titanomagnetite grains from the centre of each dyke. Percentage of total cation content given with standard deviations based on the N analyses performed.

DYKE	N	Mg	Al	Si	Ti	Cr	Mn	Fe
SA	6	0.4±0.3	5.6±1.0	1.3±0.5	17.6±1.0	0.2±0.1	0.8±0.1	74.1±1.1
UH	7	0.4±0.3	6.8±0.9	1.3±0.5	18.3±0.9	0.1±0.1	1.5±0.5	71.7±1.2
ST	5	1.7±0.4	6.2±0.6	1.0±0.3	9.1±1.1	0.2±0.2	1.3±0.7	80.5±2.4

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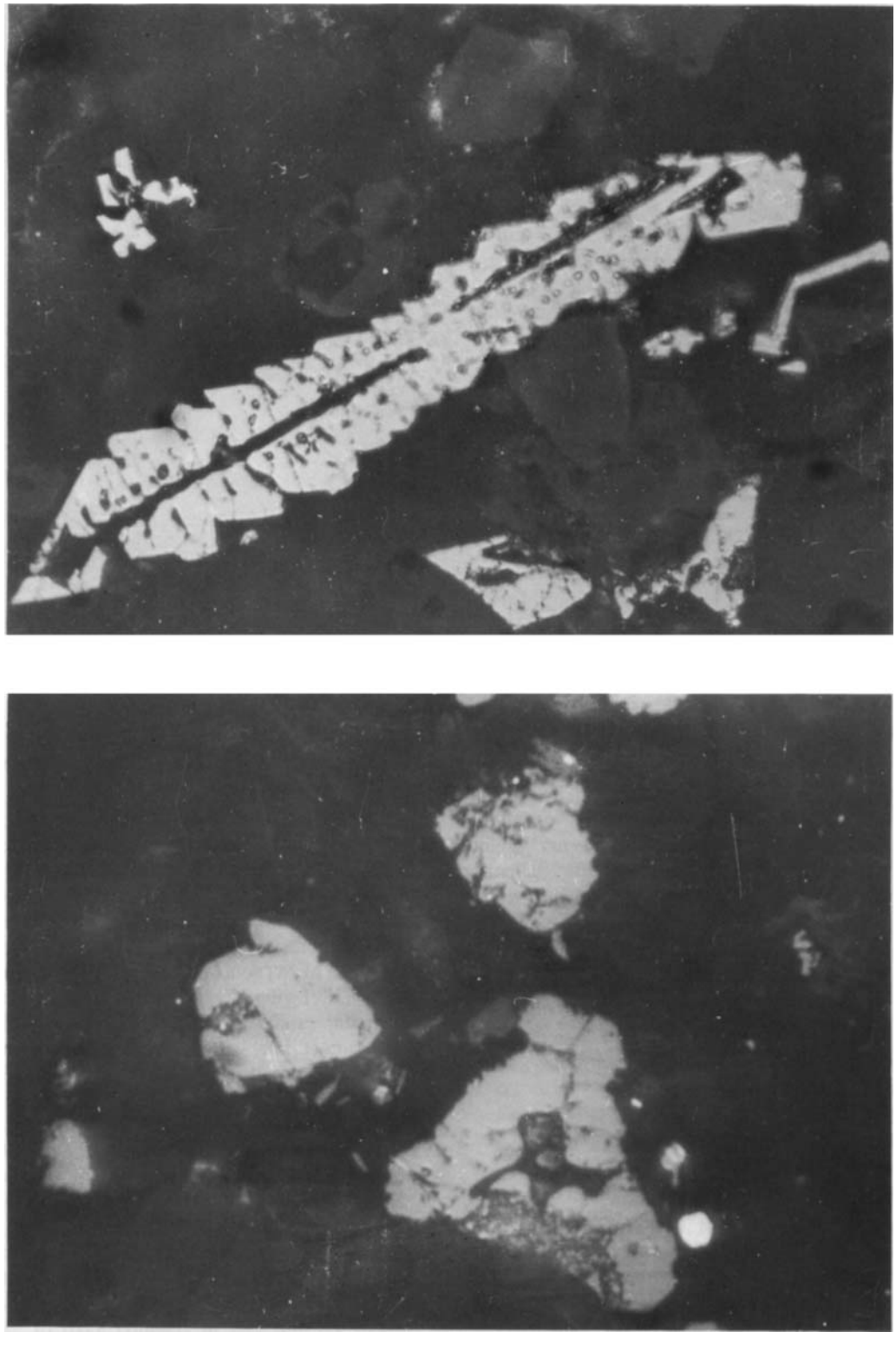
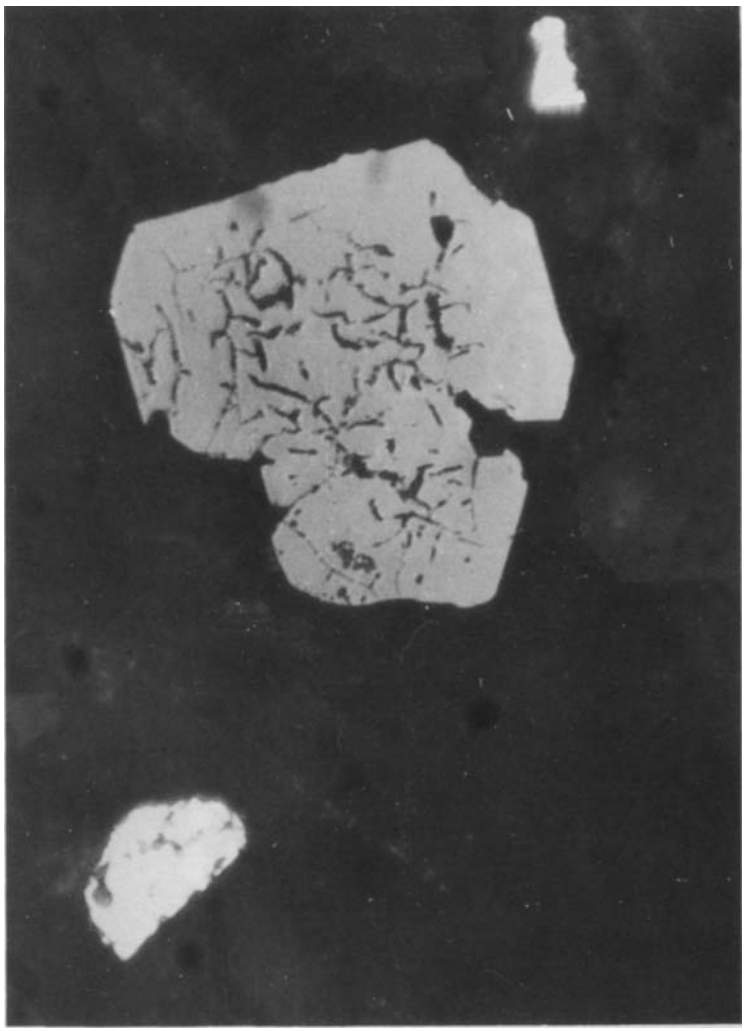


Figure 1. Reflected light micrographs of SA (top), UH (middle) and ST (bottom) titanomagnetite grains. Frame widths are 105, 60 and 105 μm respectively. Smaller bright gains are pyrite.

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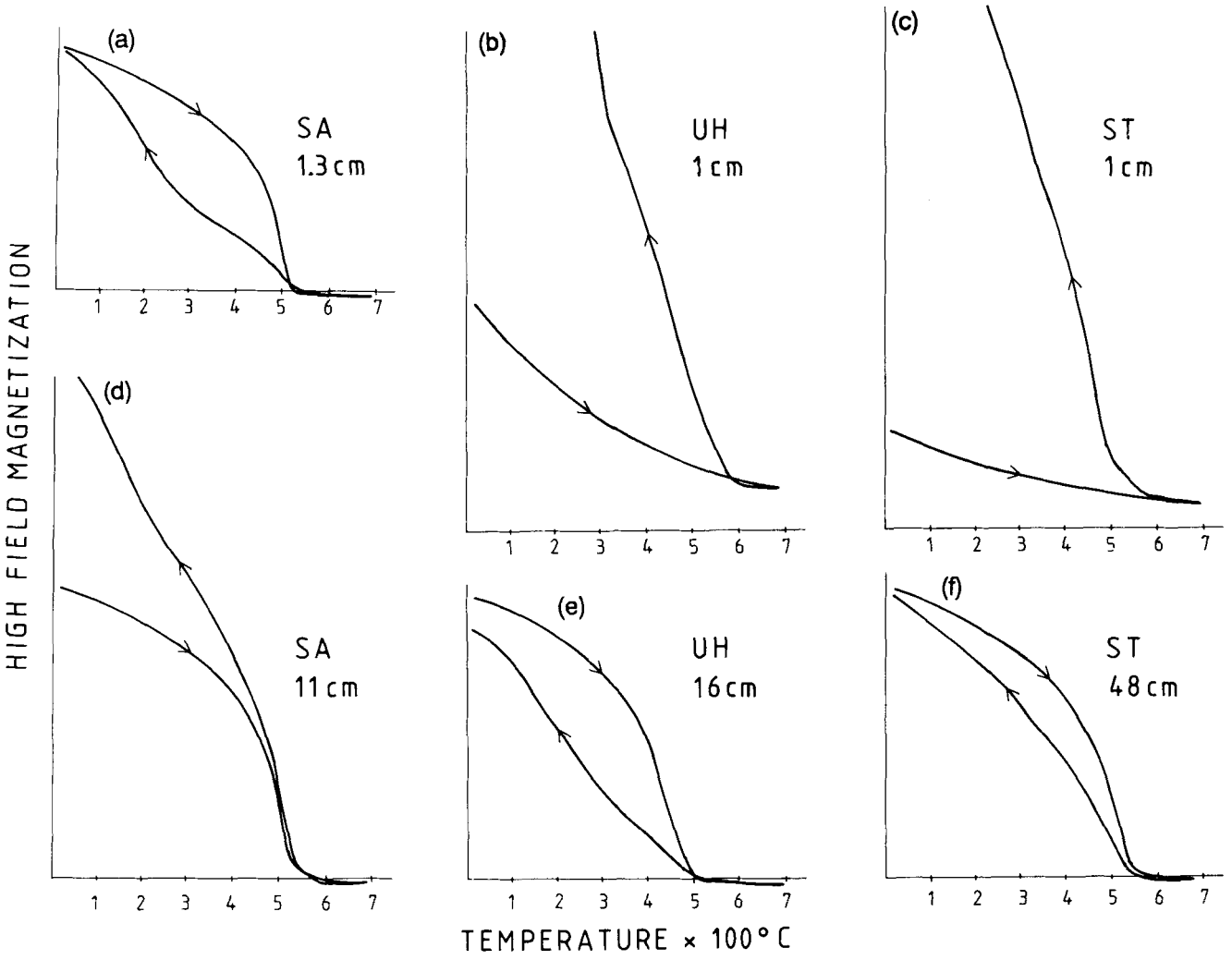


Figure 2. Thermomagnetic curves for SA, UH and ST dykes. The distance from the nearest dyke margin is indicated for each curve.

cycle to 700 °C of 1 hour, and a field of 0.6 T. T_C values during heating do not vary systematically from edge to centre of any of the dykes, with mean values of 522 ± 5 °C, 510 ± 8 °C and 546 ± 3 °C for the SA, UH and ST dykes respectively. The highest T_c is associated with the lowest Ti-content, but all dykes show T_c values substantially higher than expected for unaltered stoichiometric ATM of the bulk compositions listed above (Özdemir & O'Reilly 1981). This is consistent with a cation-deficient initial phase, but may also in part relate to partitioning into Fe- and Ti-enriched regions on a scale not visible under the optical microscope. Cation-deficient phases are likely to invert to a mixture of titanomagnetite and rhombohedral phases well below 500 °C (Readman & O'Reilly 1970; Özdemir & O'Reilly 1981; Özdemir 1987) relating the observed Curie temperatures to secondary titanomagnetite rather than the initial magnetic phase. Cycling of thermomagnetic curves to maximum temperatures around 400 °C confirm this inversion, which may also be partly responsible for the more concave shape of cooling curves (Figs 2a, e and f).

Specimens close to dyke contacts in UH and ST dykes (Figs 2b and c) show anomalous behaviour, with heating curves dominated by a paramagnetic (1/T) contribution. Pure magnetite (T_c = 580–590 °C) is formed on cooling, leading to a large increase in magnetic moment. Some cooling curves show Curie points around 325 °C, in particular after heating to above 700 °C, which are attributed to the formation of pyrrhotite.

4 ROCK MAGNETIC PROFILES AND REMANENCE

4.1 Bulk magnetic and hysteresis properties

Remanence was measured on a Digico spinner magnetometer, and bulk susceptibility and AMS on a KLY-2 low-field induction bridge. Bulk susceptibility and NRM intensity profiles over the dykes are shown together with a modified Koenigsberger ratio ($Q' = 4\pi J_{NRM}/1000K$) in Fig. 3, and mean values are summarized in Table 2.

Table 2. Bulk susceptibility, NRM intensity and modified Koenigsberger ratio ($Q' = 4\pi J_{NRM}/1000K$) for each dyke, given with standard deviations for the N specimens measured.

DYKE	N	K (m^3/kg)*10^{-7}	J_{NRM} (Am^2/kg)*10^{-4}	Q'
SA	34	105 ± 14	4.6 ± 3.1	0.59 ± 0.44
UH	46	61 ± 25	3.9 ± 2.0	0.93 ± 0.53
ST	68	74 ± 19	0.9 ± 0.3	0.15 ± 0.04

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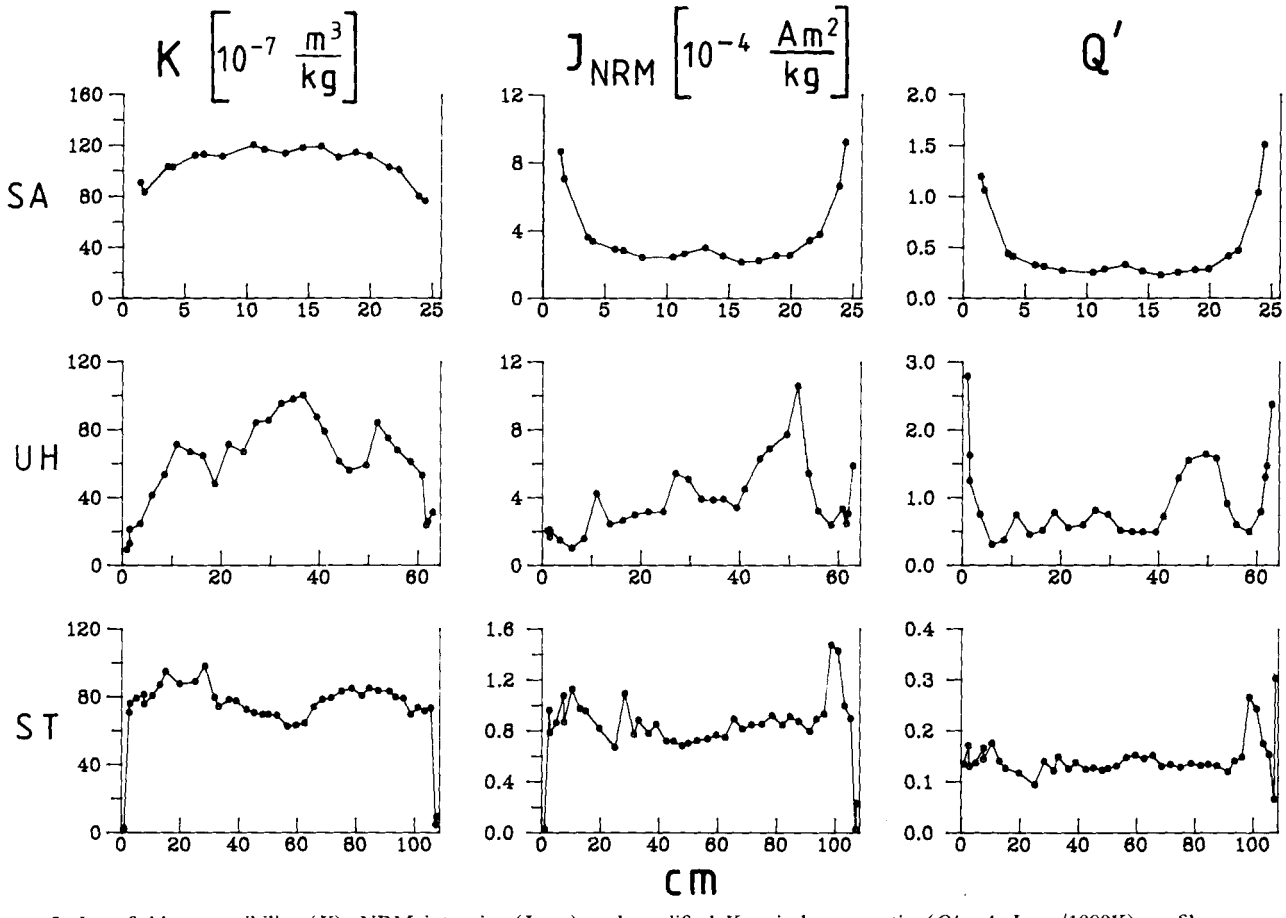


Figure 3. Low-field susceptibility (K), NRM intensity (J_{NRM}) and modified Koenigsberger ratio ($Q' = 4\pi J_{NRM}/1000K$) profiles across SA (top), UH (middle) and ST (bottom) dykes. Horizontal scale in cm from western dyke contact, with the vertical axis on the right of each individual plot indicating the Eastern dyke contact. To avoid possible surface oxidation effects, only the lowermost specimen from each drill core was used to generate the profiles.

All three dykes show symmetric margin-to-margin variations. The SA dyke gives the smoothest profiles, with remanence maxima and susceptibility minima combining to give a maximum in magnetic stability at dyke margins, in accord with the findings of Petersen (1976). A similar edge effect is evident in the UH dyke, but local maxima, possibly related to differential low-temperature oxidation, complicate the central parts of the profiles.

The broadest (ST) dyke differs by showing a very large drop in NRM intensity as well as susceptibility in a thin zone at the edges. Q' does not vary systematically, suggesting that grain size is not responsible for the edge effect in this dyke. The mean Q' value (Table 2) is an order of magnitude below that of the other dykes, indicating larger, more unstable remanence carriers.

Variations due to grain size may be separated from compositional and concentration effects by means of hysteresis measurements. Hysteresis curves were obtained on a Molspin vibrating sample magnetometer. The peak field of 1.14 T is sufficient to saturate TM and titanomaghemite grains of all sizes, but not fine-grained haematite. Hence, some uncertainty applies to the determination of hysteresis parameters in haematite-bearing specimens.

Profiles of saturation magnetization (J_s) coercive force (H_c) and the ratio of saturation remanence to saturation magnetization (J_{rs}/J_s) are shown in Fig. 4, and mean values are summarized in Table 3. Consistently high Curie points (Section 3) indicate a relatively uniform chemical composition of magnetic phases across the dykes, leaving variations

Table 3. Hysteresis properties. Initial values and values after 400 °C and 600 °C heating for each dyke, with standard deviations for the N specimens measured. K = bulk susceptibility; J_s = saturation magnetization; J_{rs}/J_s = ratio of saturation remanence to saturation magnetization; H_c = coercive force; H_{cr}/H_c = ratio of remanent coercive force to coercive force.

DYKE	N	K (m^3/kg) $*10^{-7}$	J_s (Am^2/kg)	J_{rs}/J_s	H_c (mT)	H_{cr}/H_c
			INITIAL VALUES			
SA	24	101±21	0.62±0.10	0.16±0.02	3.4±1.2	6.8±1.6
UH	10	52±30	0.40±0.25	0.17±0.03	5.4±1.7	4.6±1.0
ST	12	56±32	0.44±0.25	0.12±0.05	3.5±1.7	9.9±2.7
			AFTER 400°C HEATING			
SA400	24	96±20	0.60±0.10	0.17±0.01	3.4±1.1	5.4±0.9
UH400	10	50±31	0.39±0.25	0.17±0.02	5.3±1.2	4.8±0.7
ST400	12	57±32	0.43±0.25	0.13±0.03	3.5±1.8	9.5±3.3
			AFTER 600°C HEATING			
SA600	24	102±32	1.02±0.29	0.31±0.03	16.9±3.9	2.3±0.2
UH600	10	29±16	0.33±0.19	0.36±0.03	21.8±6.0	2.1±0.2
ST600	12	54±31	0.50±0.25	0.25±0.07	13.9±7.7	3.2±0.8

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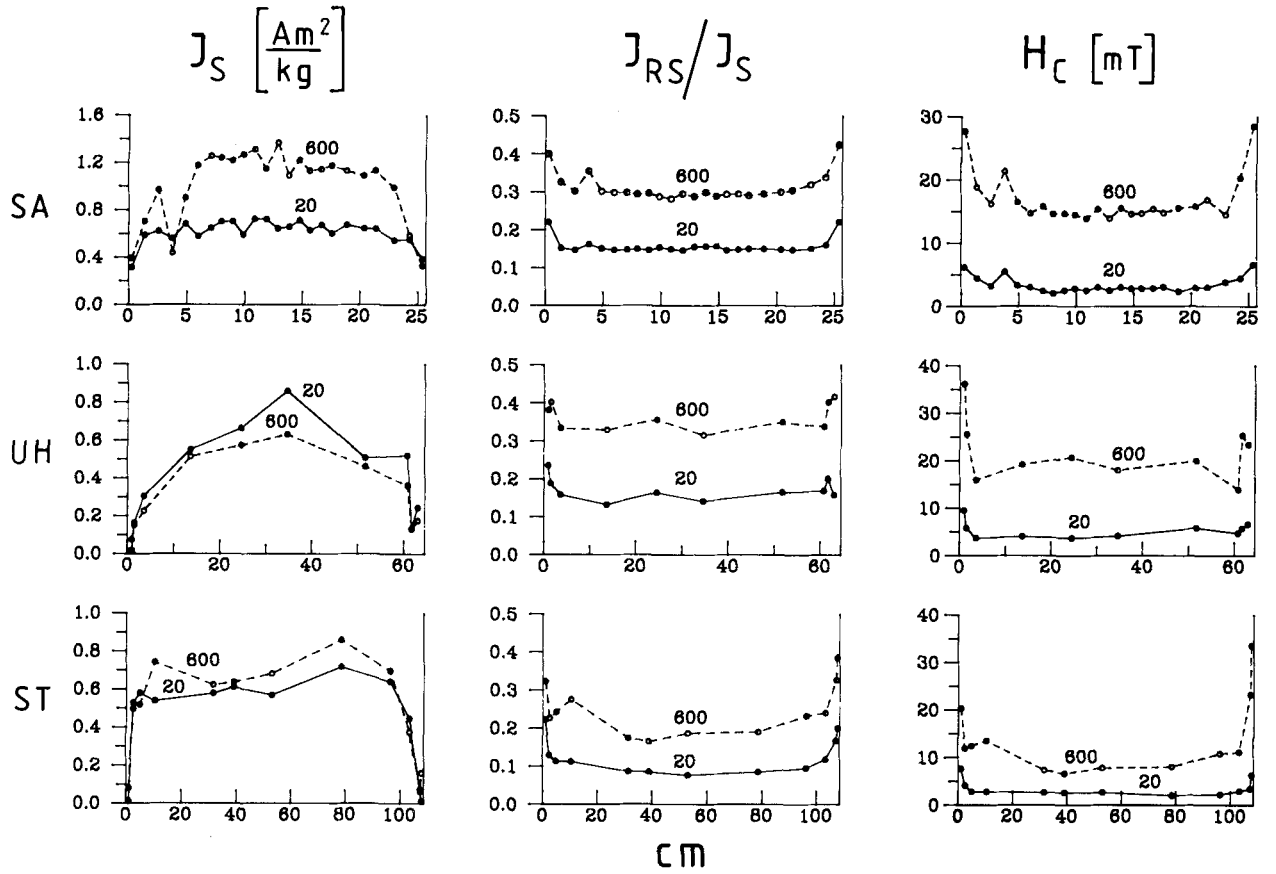


Figure 4. Profiles of saturation magnetization (J_s), ratio of saturation remanence to saturation magnetization (J_{rs}/J_s) and coercive force (H_c) before (marked '20') and after heating specimens to 600 °C.

in TM concentration as the main controlling influence on J_s. The profiles therefore indicate a significant lowering of TM concentration near dyke margins. The very large drop in J_{NRM} and K (Fig. 3) at dyke margins for the broadest dyke is mirrored by J_s, suggesting a drop in TM concentration to less than 3 per cent of the mean value.

The ratio J_{rs}/J_s is commonly regarded as the best measure of magnetic domain state and grain size, ranging from 0.5 for uniaxial SD grains down to <0.02 for true MD magnetite (Dunlop 1986). Increasing Ti content in TM raises the MD/PSD transition value to 0.1 (Dunlop 1981).

The three dykes all fall within the PSD range, with the central part of the ST dyke close to true MD values. A decrease in magnetic grain size near dyke margins is evident in all dykes, but only the broadest dyke, which has the least central variation in K and J_{NRM}, shows a small consistent variation across the entire dyke width.

4.2 Directions and stability of remanence

Thermal or AF demagnetization was performed on a total of 55 specimens from the three dykes, utilizing a Schonstedt TSD-1 furnace, and a two-axis tumbler with a maximum field of 100 mT. NRM stability (Fig. 5) does not vary systematically across the SA and UH dykes. The remanence in the ST dyke is extremely soft except in the low-intensity samples near dyke contacts, where median destructive fields rise from below 5 mT to approximately 30 mT.

Characteristic remanence directions and NRM directions ae summarized in Table 4 and Figs 6 and 7. The SA and UH dykes both have two remanence components, with the softer parallel to the present-day Earth field for the UH dyke, and more erratic for the SA dyke. Demagnetization to 300 °C or 7 mT, leaves a stable well-defined remanence (Figs 7b and d), with maximum blocking temperatures in excess of 500 °C. Mean directions (Table 4) define pole positions situated between the late Triassic pole position for Europe (Jowett & Pearce 1987; Van der Voo 1990) and the *A'* pole of Løvlie (1981).

Most specimens from the ST dyke contain a single low-stability component (Fig. 6c), parallel to the present-day field. Maximum blocking temperatures around 300 °C are in reasonable accord with a VRM acquired during the last 730 000 yr in the Brunhes normal polarity epoch (Pullaiah *et al.* 1975; Middleton & Schmidt 1982). The only specimens to retain a second, harder magnetization are those in the anomalous zone less than 4 cm from either dyke margin (Fig. 6d). These 'edge components' do not define a consistent mean direction (Fig. 8), but a best-fit plane through them coincides with the plane defined by the VRM (or present-day field) and the axis of stable remanence for the two thinner dykes, suggesting a partial retention of the older (Triassic?) magnetization. This inference is, however, highly speculative, since all the studied stable dykes in the Sunnhordland complex are of normal polarity (Løvlie 1981), while the suggested connection would require a reversed component.

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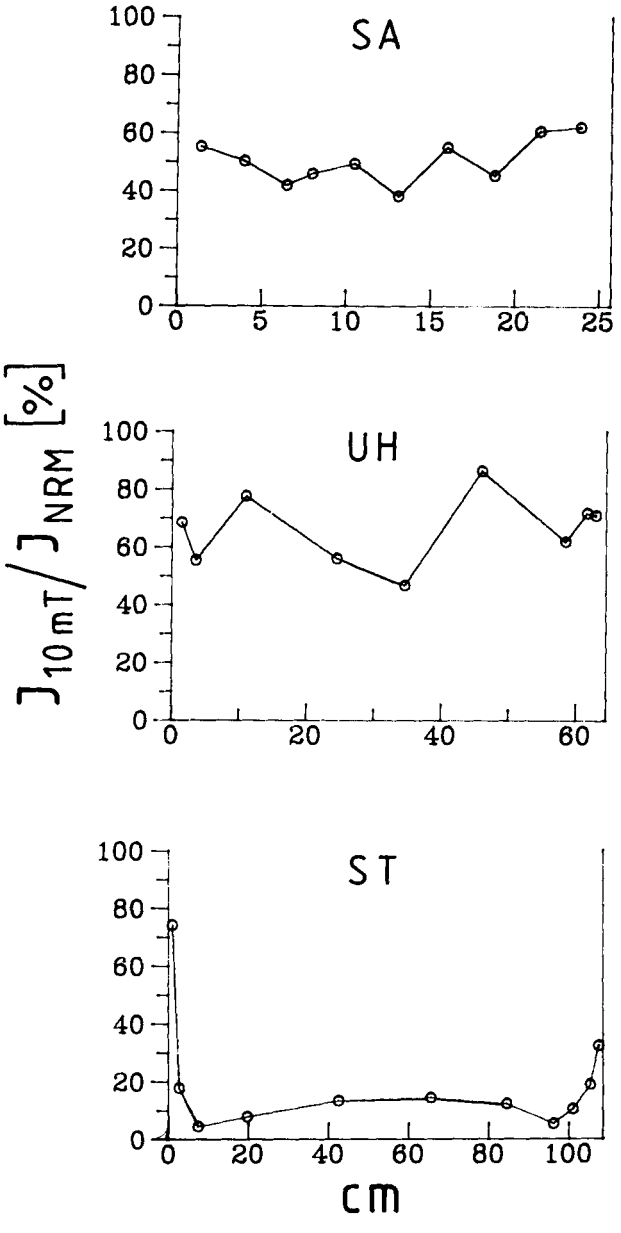


Figure 5. Profiles of residual fraction of NRM intensity after 10 mT AF demagnetization.

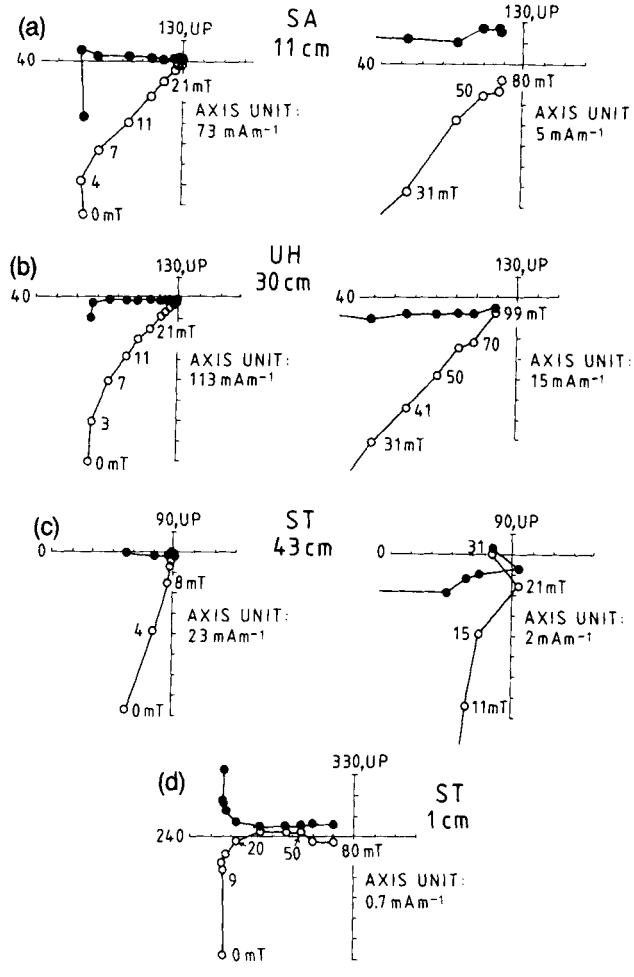


Figure 6. Representative AF demagnetization results from the central part of SA (a), UH (b) and ST (c) dykes, and from the low-intensity zone close to the margin of the ST dyke (d), shown in optimal orthogonal vector projections. Open and closed symbols denote vertical and horizontal projections respectively. The distance from the nearest dyke margin is indicated for each specimen. Right-hand side diagrams are expansions of the high-blocking portions of the magnetization.

Table 4. Directional statistics for NRM directions and the principal remanence component in each dyke. N = number of specimens, with bracketed value for the principal components denoting total number of demagnetized specimens; D/I = mean directions; k = Fisher precision parameter; α_{95} = 95 per cent confidence radius of the mean. N values for the NRM are slightly lower than in Tables 2 and 5, due to the exclusion of unoriented cores. Note that principal component directions should not be considered palaeomagnetically reliable, owing to the limited number of demagnetized specimens, and the use of a single hand sample in the SA dyke.

	NRM				PRINCIPAL COMPONENT			
DYKE	N	D/I	k	α_{95}	N	D/I	k	α_{95}
SA	32	059/57	7	10.2	16 {18}	041/50	367	1.9
UH	40	035/54	176	1.7	15 {15}	040/48	233	2.5
ST	66	003/72	41	2.8	22 {22}	001/73	58	4.1

4.3 Thermally induced changes

In order to investigate the effects of secondary chemical alterations, bulk magnetic and hysteresis properties were measured after progressive step heating to a maximum of 600 °C. Heating runs above 600 °C were unsuccessful, because specimens tended to split up along pre-existing microcracks.

With the exception of remanent intensity, there is little change in magnetic properties below 400 °C. Between 400 °C and 600 °C, H_c and J_{rs}/J_s rise sharply (Fig. 4 and Table 3), implying a reduction in magnetic grain size. Maximum J_{rs}/J_s values of around 0.4 indicate grains near the true SD range. A fall in K between 400 °C and 550 °C (Fig. 9) is also attributed to grain-size reduction. Susceptibility rises again between 550 °C and 600 °C, possibly reflecting the formation of secondary magnetite seen in the J_s–T curves (Figs 2b and c).

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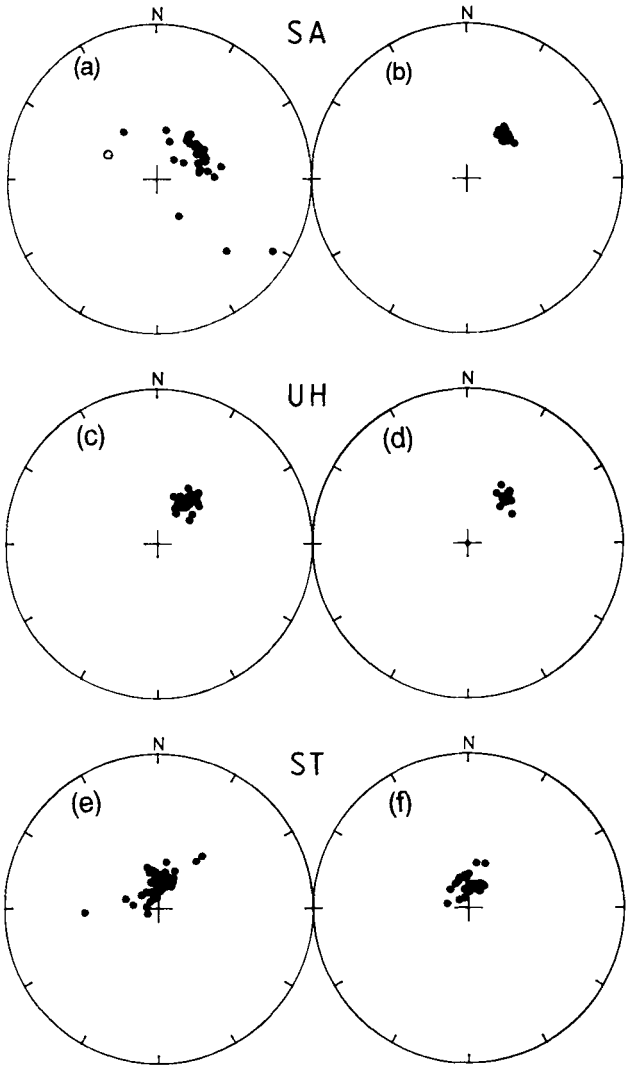


Figure 7. NRM directions (left) and characteristic remanence components (right) for each dyke, shown in stereographic projections. Closed and open symbols denote positive and negative inclinations respectively.

5 AMS

AMS was measured initially on all specimens, and after progressive step heating to 200, 400, 500 and 600 °C for selected specimens from each dyke. The AMS parameters used conform to suggestions for standardization put forward by Ellwood, Hrouda & Wagner (1988), and are defined using K_1, K_2 and K_3 to denote maximum, intermediate and minimum susceptibilities respectively. Mean values are presented in Table 5. The degree of anisotropy $P = K_1/K_3$ (Nagata 1961) (Fig. 10), magnetic lineation $L = K_1/K_2$ (Balsley & Buddington 1960), and magnetic foliation $F = K_2/K_3$ (Stacey, Joplin & Lindsay 1960) all vary symmetrically across the SA dyke, with maximum values near dyke edges, while the SU and ST dykes show no obvious trends. The ellipsoid shape factor V of Graham (1966) [defined by $\sin^2 V = (K_2 - K_3)/(K_1 - K_3)$] shows that both prolate ($V < 45°$) and oblate ($V > 45°$) ellipsoid shapes are present in all three dykes. The thinnest again shows systematic variations, with strongly oblate shapes near both dyke margins.

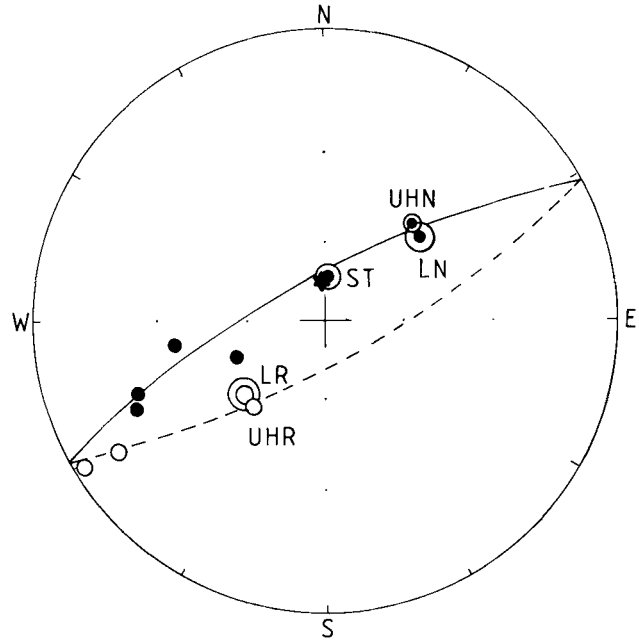


Figure 8. Highest stability components for six of seven demagnetized specimens closer than 4 cm from either margin of the ST dyke, shown with best-fit plane. UHN, UHR, LN and LR denote normal and reversed mean directions from the UH dyke and Løvlie (1981) respectively, shown with α_{95} confidence circles. Also shown is the mean of the dominant low-stability component in the ST dyke, and the present-day dipole field direction (star).

AMS directions show internal consistency within each of the two broadest dykes (Fig. 11), but differ between them, with maximum susceptibility axes perpendicular to dyke walls in the UH dyke, and parallel in the ST dyke. The SA dyke shows systematic directional variations between margins and centre, concurrent with the variations in ellipsoid parameters.

Specimens drilled through the country rock between 10 and 15 cm from dyke contacts in the UH dyke (Fig. 11b), and through the actual contacts of the ST dyke (Fig. 11c), show internally consistent directions differing from dyke directions. This 'contact test' indicates that initial anisotropy in the dyke and/or the country rock is not totally overprinted by later tectonic stress.

Heat treatment leads to a general reduction of anisotropy in the three dykes (Fig. 12). After heatings below 400 °C, this may be due to the removal of 'secondary' anisotropies related to tectonic stress (Park, Tanczyk & Desbarats 1988) and low-temperature oxidation. Above 400 °C alteration of magnetic minerals and formulation of new magnetic phases in a low-stress environment further contributes to a lowering of anisotropy. Symmetric AMS profiles emerge on heating from the initially confusing picture in the two broadest dykes (Fig. 12), supporting the notion of a primary dyke-emplacement effect overlain by spurious secondary AMS components.

The AMS directions of the two broadest dykes are not significantly changed by heating, except when the lowering of anisotropy leads to near-isotropic specimens with poor axis definition. Most specimens from the SA dyke, however, undergo large axis shifts, with K_3 directions changing by 90° (Figs 13a and b). Only the specimens closest to dyke

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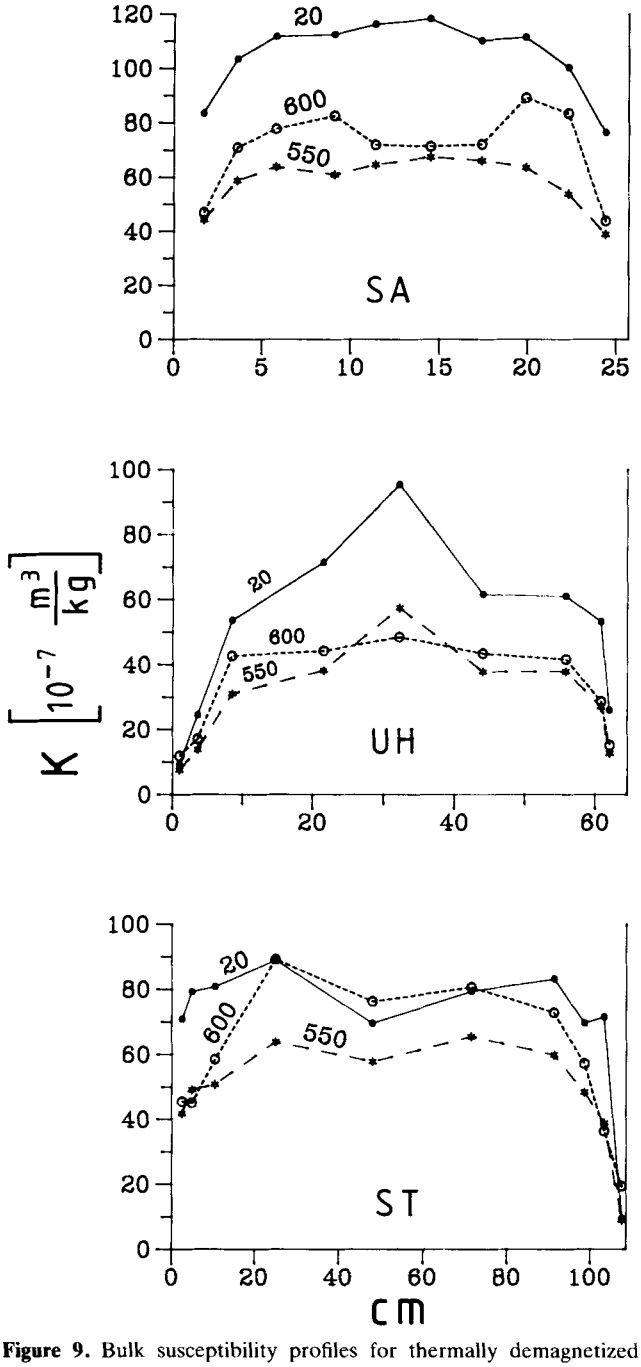


Figure 9. Bulk susceptibility profiles for thermally demagnetized specimens prior to laboratory heating (marked '20'), and after demagnetization to 550 °C and 600 °C respectively.

margins retain original axis directions, leading to a much-improved definition of mean AMS axes after heating to 600 °C, despite the general lowering of anisotropy. Ellipsoid shapes also change, as illustrated in Fig. 13(c).

Improvement in the grouping of AMS axes after heating has been reported previously for both igneous, metamorphic and sedimentary rocks (Jelenska & Kadzialko-Hofmokl 1990). Park *et al.* (1988), who studied Proterozoic Canadian diabase dykes, reported the emergence of an AMS related to dyke emplacement after a secondary anisotropy, presumably related to tectonic stress, had been removed by thermal demagnetization to 640 °C. They obtained the same

Table 5. AMS parameters with standard deviations for the N specimens measured. $P = K_1/K_3$; $L = K_1/K_2$; $F = K_2/K_3$. The ellipsoid shape factor V is defined by $\sin^2 V = (K_2 - K_3)/(K_1 - K_3)$, with values above (below) 45° indicating oblate (prolate) ellipsoids.

DYKE	N	Anisotropy P	Lineation L	Foliation F	Shape factor V
SA	34	1.026±0.009	1.012±0.005	1.014±0.009	45±13
UH	46	1.049±0.022	1.032±0.020	1.017±0.008	37±12
ST	68	1.016±0.006	1.010±0.005	1.005±0.003	36±13

changes in AMS directions by AF demagnetization to between 70 and 100 mT. AF demagnetization to 80 mT for nine specimens from the SA dyke, however, had no significant effect on axis orientations.

The reduction of magnetic grain size on heating (Fig. 4) offers another possible explanation for the changes in AMS portrayed in Fig. 13. If the grains were controlled by

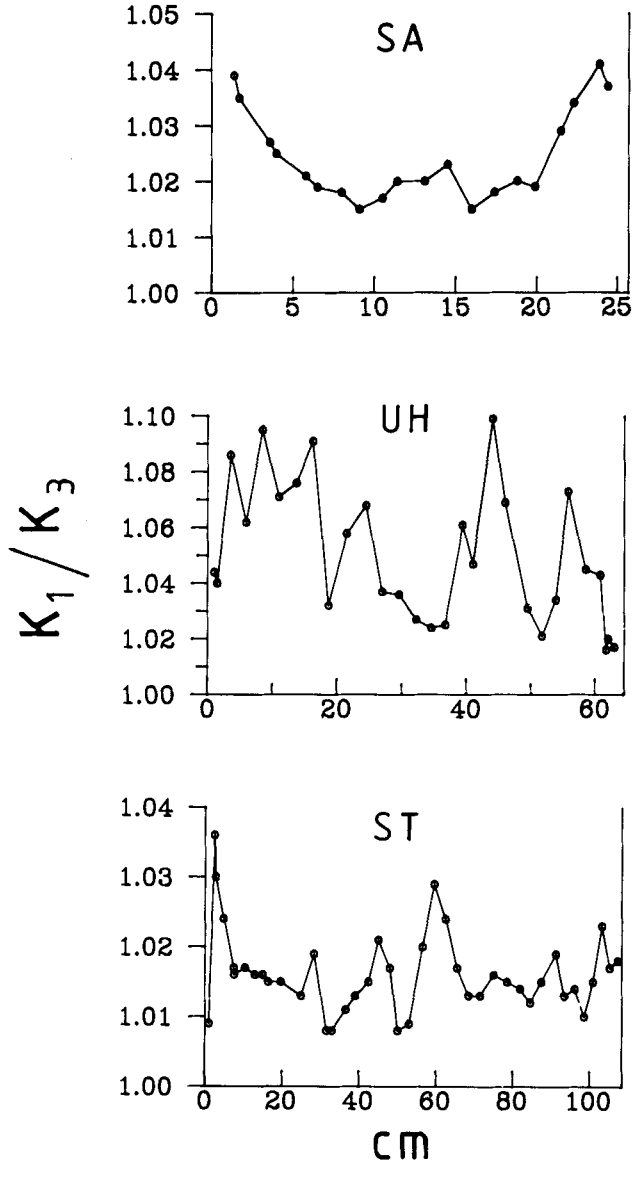


Figure 10. Profiles of the degree of anisotropy $P = K_1/K_3$, utilizing the lowermost specimens from each individual drill core.

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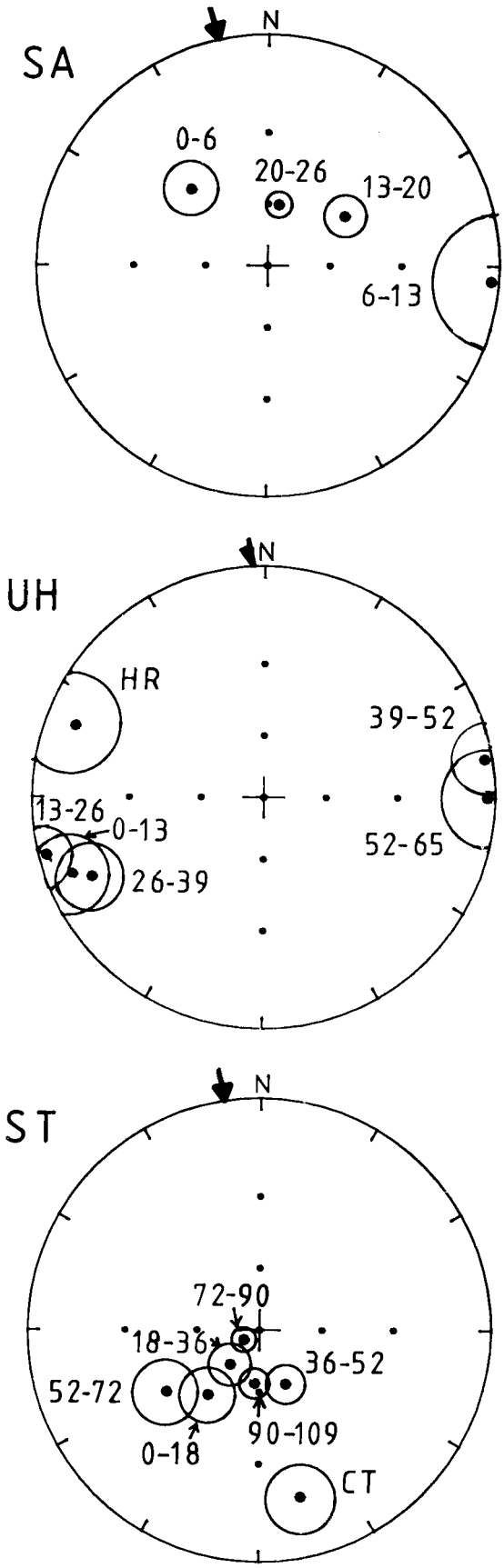


Figure 11. Lower hemisphere plots of changes in direction and scatter of maximum susceptibility axes ($\mathbf{K}_1$) traversing across the dykes. Circles represent α_{95} confidence limits for the mean of all measured specimens falling between the indicated minimum and maximum distances (in cm) from the western dyke margins. The circles marked 'HR' in the middle diagram represents four specimens drilled in the host rock adjacent to the UH dyke, and the circle marked 'CT' in the bottom diagram represents four specimens drilled through the actual contacts of the ST dyke. Strike directions for the dykes are indicated by arrows.

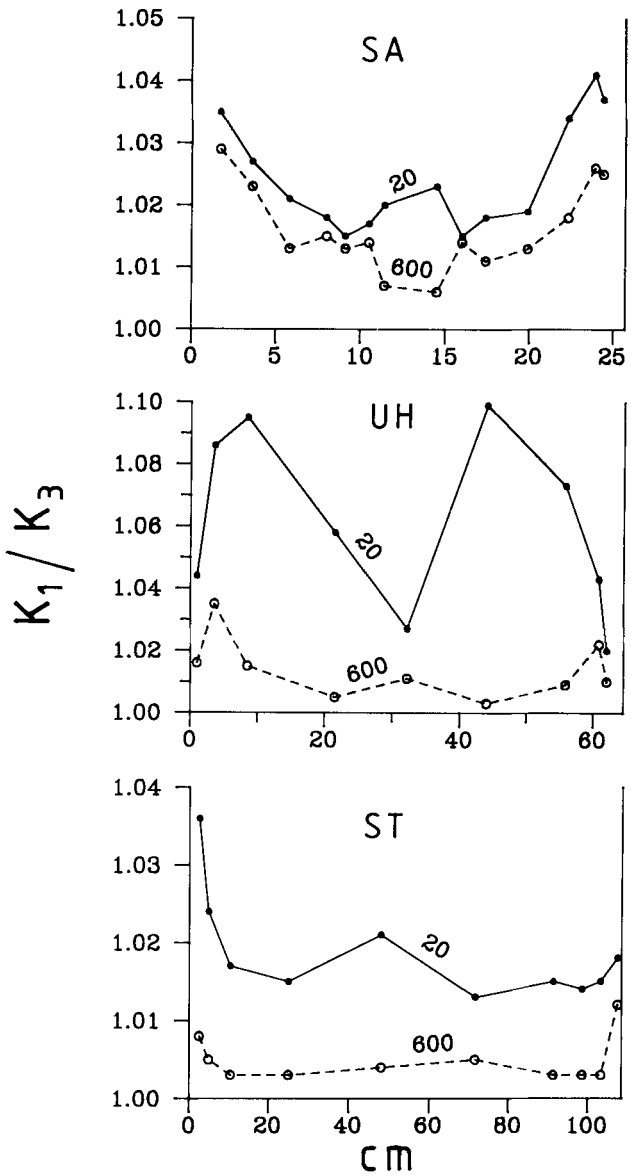


Figure 12. Profiles of the degree of anisotropy $P = (K_1/K_3)$ before ('20') and after heating selected specimens to 600 °C.

uniaxial shape anisotropy, a transition from PSD to SD behaviour would change the longest axis of the grains from a maximum to a minimum axis of susceptibility (Stephenson, Sadikun & Potter 1986; Potter & Stephenson 1988). This mechanism could explain both the change in ellipsoid shape and minimum axis orientation, and is consistent with the observation that only specimens near dyke edges, which have the smallest initial grain sizes, show no change in axis orientations. To test this possibility, anisotropic isothermal remanent magnetization (AIRM), which does not show the

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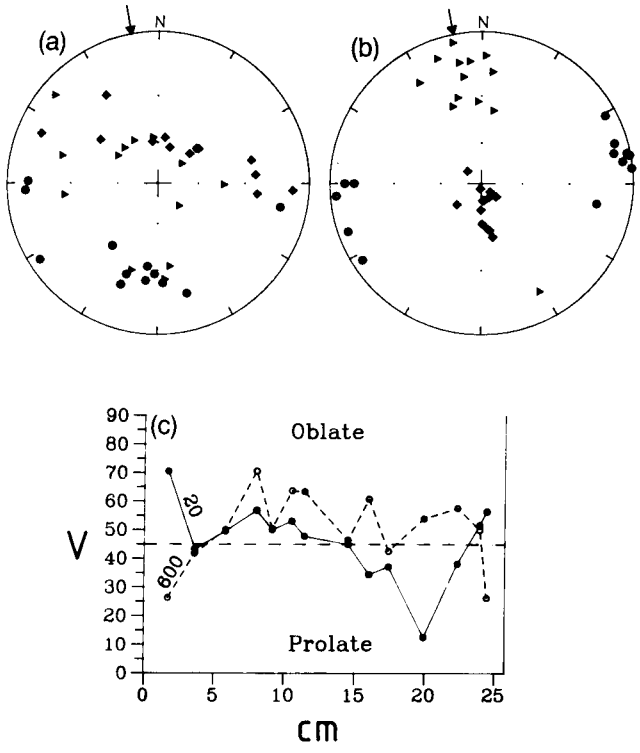


Figure 13. Heat-induced changes in AMS for the SA dyke. (a) and (b): lower hemisphere plots of AMS principal axis directions of individual specimens before (a) and after (b) heating to 600 °C. Maximum, intermediate and minimum directions are denoted by squares, triangles and circles respectively. Arrows indicate dyke strike. (c): Profiles of the ellipsoid shape factor V before and after heating to 600 °C.

same axis inversion as AMS at the SD transition (Stephenson *et al.* 1986; Potter & Stephenson 1988) was used. An AIRM in steady field of 50 mT was induced in four specimens from the SA dyke before and after heating to 600 °C, using the method described by Stephenson *et al.* (1986). AIRM minimum axes coincided with AMS minimum axes both before and after heating, with both anisotropies displaying large, but identical directional shifts. This seems to rule out any explanation in terms of PSD–SD transitions.

6 DISCUSSION

The SA, UH and ST dykes are thinner and of much greater age than those studied by Petersen (1976) and Smith & Prevot (1977). In contrast to these studies, the symmetry of magnetic profiles across the Sunnhordland dykes is not solely due to grain-size variations, but also a lowering of titanomagnetite concentration towards dyke margins. A decrease in TM concentration towards cooling surfaces has also been found in pillow basalts (Marshall & Cox 1971) and lava flows (Schoenharting & Ghisler 1982). Such a concentrational dependence helps explain that neither laboratory reheating nor low-temperature secondary oxidation homogenizes the magnetic properties across the dykes.

The origin of the large difference in remanent stability between dykes, is suggested to be the difference in bulk titanium content in TM grains (Section 3). Secondary partitioning of TM grains into Fe- and Ti- enriched regions has taken place in all three dykes, accounting for the high observed Curie points. This would lead to a significant reduction in effective magnetic grain size in the two thinner dykes, which have Fe/Ti ratios comparable to sea-floor basalts, leaving stable magnetic carriers of PSD size. Only half this amount of titanium is present in TM grains in the ST dyke, leading to less secondary partitioning, and leaving larger iron-enriched regions closer to the MD-size range.

The fine-grained zones near dyke margins carry the most stable remanence, but are at the same time more vulnerable to secondary CRM resetting (Halls 1991). A large surface-to-volume ratio in magnetic grains, combined with easier access of water and oxygen due to marginal cracks, may lead to a higher oxidation state, as observed in the ST dyke. Moderate single-phase oxidation of small TM grains does not change remanence directions, but if oxidation proceeds too far, with formation of new rhombohedral magnetic phases, directional resetting will occur (Dunlop 1990).

Formation of secondary magnetic minerals from non-magnetic matrix material is also most likely to occur near dyke margins, and will have a relatively larger impact on bulk magnetic properties due to the lower concentration of magnetic material. This is demonstrated by the relative change of susceptibility on heating for the small specimens used in hysteresis measurements (Fig. 14). Due to a larger surface-to-volume ratio, and sampling in closer proximity to dyke margins, these profiles show an edge effect not readily seen for the larger specimens in Fig. 9. The large increase in K at the margins of the ST dyke is **not** due to grain-size reductions into the SP size range, since J_{rs} shows a comparable relative increase, but rather the formation of secondary pure magnetite demonstrated in Fig. 2(c). Optical microscope observations and EDA measurements suggest that some secondary magnetite has formed near dyke edges prior to laboratory heating. Smaller grain sizes in the primary TM phase are therefore unlikely to be the sole cause of the sharp rise in remanent stability near the margins of the ST dyke.

The relationship between AMS axis directions and dyke geometry (Khan 1962; Ellwood 1978, 1979; Knight & Walker 1988; Park *et al.* 1988) is ambiguous. Individual AMS components may relate to magma-flow directions, cooling stress, and tectonic stress associated with dyke emplacement, rendering any predictions of 'expected' initial AMS directions uncertain. Computer modelling by Dawson & Hargraves (1985) suggested that a dyke strained by simple shear and solidifying during magma flow, should have $\mathbf{K}_1$ axes parallel to the flow directions, and $\mathbf{K}_3$ axes perpendicular to dyke walls. They further proposed that $\mathbf{K}_1$ axes perpendicular to dyke walls could arise from crystals growing out into the magma from the cooler dyke walls when no flow is taking place. The flow scenario agrees well with the SA dyke directions after heat treatment if a near-vertical magma emplacement direction is assumed, while the UH dyke fits the second case. The broadest dyke has $\mathbf{K}_1$ directions similar to the SA dyke, but a large spread in $\mathbf{K}_3$ and $\mathbf{K}_2$ directions. This spread relates mainly to poor axis definition due to low-magnetic foliation (Table 5), with four of 68 measured specimens showing no difference in length between the two shorter axes at the 95 per cent confidence level (Jelinek 1977). Interpretation of the ST

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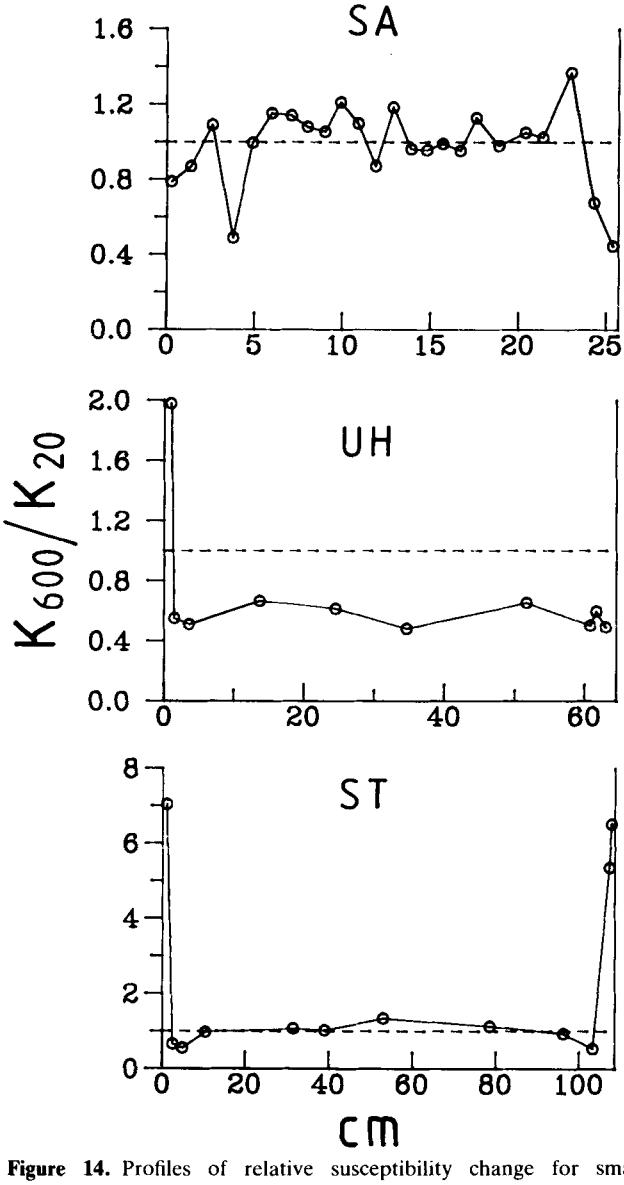


Figure 14. Profiles of relative susceptibility change for small specimens used in hysteresis experiments after heating to 600 °C.

anisotropy in terms of the above model is therefore problematic, but a primary flow-controlled magnetic fabric remains a possibility.

Increasing anisotropy towards dyke edges, which is obvious in all three dykes after 600 °C heating (Fig. 12), may relate to solidification of the dykes, which takes place from the margins inward as the magma cools. In the central parts a fluid or plastic state is retained longer, enabling a more efficient release of contraction stress during cooling and partial randomization of AMS by small-scale convection.

7 CONCLUSIONS

The conclusions of this study may be summarized as follows:

—Symmetric variations in magnetic properties across the three dykes are caused by varying titanomagnetite concentration as well as grain size.

—The symmetric variation is modified, but not destroyed, by low temperature and heat-induced oxidation.

—Grains near dyke edges carry the most stable remanent magnetization due to smaller initial grains sizes, but are also most vulnerable to secondary low-temperature oxidation and CRM overprinting.

—Despite roughly similar optical grain size and magnetic mineralogy, between dyke variations in remanent stability are larger than within dyke variations, with a soft, unstable VRM parallel to the present-day field in the broadest dyke and a stable remanence in the two thinner dykes. This is attributed to a lower initial titanium content in the TM phases of the broadest dyke, which impedes magnetic partitioning of the grains during oxidation.

—AMS directions in the dykes differ from those in the country rock. Anisotropy increases towards dyke edges, and shows a simpler pattern after heating in all three dykes. This is attributed to the removal of overprinted contributions due to low-temperature oxidation and tectonic stress, leaving a 'primary' AMS, which may reflect initial crystal growth and cooling stress.

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